Stem Cell Research 83 (2025) 103652

Contents lists available at ScienceDirect

Stem Cell Research

journal homepage: www.elsevier.com/locate/scr

Generation of induced pluripotent stem cell line from a patient with long COVID

David Wu [a,c,1], Amit Manhas [a,b,d,1], Chikage Noishiki [a,c], Dipti Tripathi [b,c], Lu Liu [a,c], Naima Turbes [a,c], Dilip Thomas [a,d], Karim Sallam [a,d], Jason T. Lee [a,b,c], Nazish Sayed [a,b,c,*]

[a] Stanford Cardiovascular Institute, Stanford University School of Medicine, CA, USA
[b] Baszucki Family Vascular Surgery Biobank, USA
[c] Division of Vascular Surgery, Department of Surgery, Stanford University School of Medicine, CA, USA
[d] Division of Cardiovascular Medicine, Stanford University School of Medicine, CA, USA

ARTICLE INFO

Keywords:
Long COVID
Induced pluripotent stem cells
Inflammation

ABSTRACT

Long COVID, or post-acute sequelae of SARS-CoV-2 infection, leads to vascular dysfunction, which contributes to the chronic multi-organ damage often seen in affected patients. Long COVID, a global health concern is associated with increased thrombotic risk, also known as COVID-19-associated coagulopathy (CAC). Here, we derived an induced pluripotent stem cell (iPSC) line from peripheral blood mononuclear cells (PBMCs) of a long COVID patient. This iPSC line showed normal morphology, maintained pluripotency, had a stable karyotype, and demonstrated the ability to differentiate into the three germ layers (ectoderm, endoderm, and mesoderm). This line provides a valuable tool for modeling long COVID and exploring mechanisms underlying multi-organ dysfunction.

Resource table

Unique stem cell lines identifier	BFVSBi005-A
Institution	Baszucki Family Vascular Surgery Biobank
Contact information of the reported cell line distributor	Dr. Nazish Sayed
Type of cell lines	iPSC
Origin	Human
Additional origin info(Applicable for human ESC or iPSC)	BFVSBi005-A; Age 60–70, Sex: Male, Ethnicity: White
Cell Source	PBMCs
Method of reprogramming	Integration-free Sendai virus expressing human OCT4, SOX2, KLF4, and c-MYC
Clonality	Clonal
Evidence of the reprogramming transgene loss	RT/q-PCR
Type of Genetic Modification	None
Associated disease	Long COVID
Gene/locus	N/A
Date archived/stock date	BFVSBi005-A: 8-4-2020
Cell line repository/bank	https://hpscreg.eu/user/cellline/edit/BFVSBi005-A
Ethical approvals	The Administrative Panel approved the generation of the lines on Human Subjects Research (IRB) under IRB #62122, "Human Induced Pluripotent Stem Cells for Studying Cardiac and Vascular Diseases."

(*continued on next column*)

(*continued*)

1. Resource utility

The iPSC line in this study was generated from one patient exhibiting severe symptoms of post-acute COVID-19, requiring surgery for arterial or venous thrombosis. This iPSC line provides a resource to model Long COVID *in vitro* for mechanistic studies and drug discovery (Table 1).

2. Resource details

Long COVID, also referred to as post-acute sequelae of SARS-CoV-2 infection (PASC), describes patients who experience three or more COVID-19 symptoms for 12 weeks or longer following the initial infection, with symptoms persisting for at least 8 weeks. These patients, regardless of their vaccination status, continue to experience lasting

* Corresponding author at: 240 Pasteur Drive, 3551 Biomedical Innovations Building, Stanford, CA 95304, USA.
E-mail address: sayedns@stanford.edu (N. Sayed).
[1] Authors contributed equally.

https://doi.org/10.1016/j.scr.2025.103652
Received 6 November 2024; Received in revised form 2 January 2025; Accepted 3 January 2025
Available online 4 January 2025
1873-5061/© 2025 The Authors. Published by Elsevier B.V. This is an open access article under the CC BY-NC-ND license (http://creativecommons.org/licenses/by-nc-nd/4.0/).

Table 1
Characterization and validation.

Classification	Test	Result	Data
Morphology	Photography Bright field	Normal	Fig. 1A
Phenotype	Qualitative analysis (Immunocytochemistry)	Positive expression of pluripotency markers: SOX2, SSEA4, and TRA-1–60	Fig. 1B
	Quantitative analysis (RT-qPCR)	mRNA expression of SOX2 and NANOG	Fig. 1D
Genotype	Karyotype: Whole genome array (KaryoStat™ Assay) Resolution 1–2 Mb	Normal karyotype: 46 XY for BFVSBi005-A	Fig. 1G
Identity	Microsatellite PCR (mPCR) or	N/A	N/A
	STR analysis	16 loci tested, 100 % matching identity	Submitted in the archive with Journal
Mutation analysis	Sequencing	N/A	N/A
	Southern Blot OR WGS	N/A	N/A
Microbiology and virology	Mycoplasma	Luminescence: Negative	Fig. 1F
Differentiation potential	Directed differentiation, Immunofluorescence staining for 2 markers per germ layer	Positive Immunofluorescence staining of three germ layer markers **Ectoderm:** PAX6, OTX2 **Endoderm:** SOX17, FOXA2, **Mesoderm:** BRACHYURY, TBX6	Fig. 1C
Donor screening	HIV 1 + 2 Hepatitis B, Hepatitis C	N/A	N/A
Genotype additional info	Blood group genotyping	N/A	N/A
	HLA tissue typing	N/A	N/A

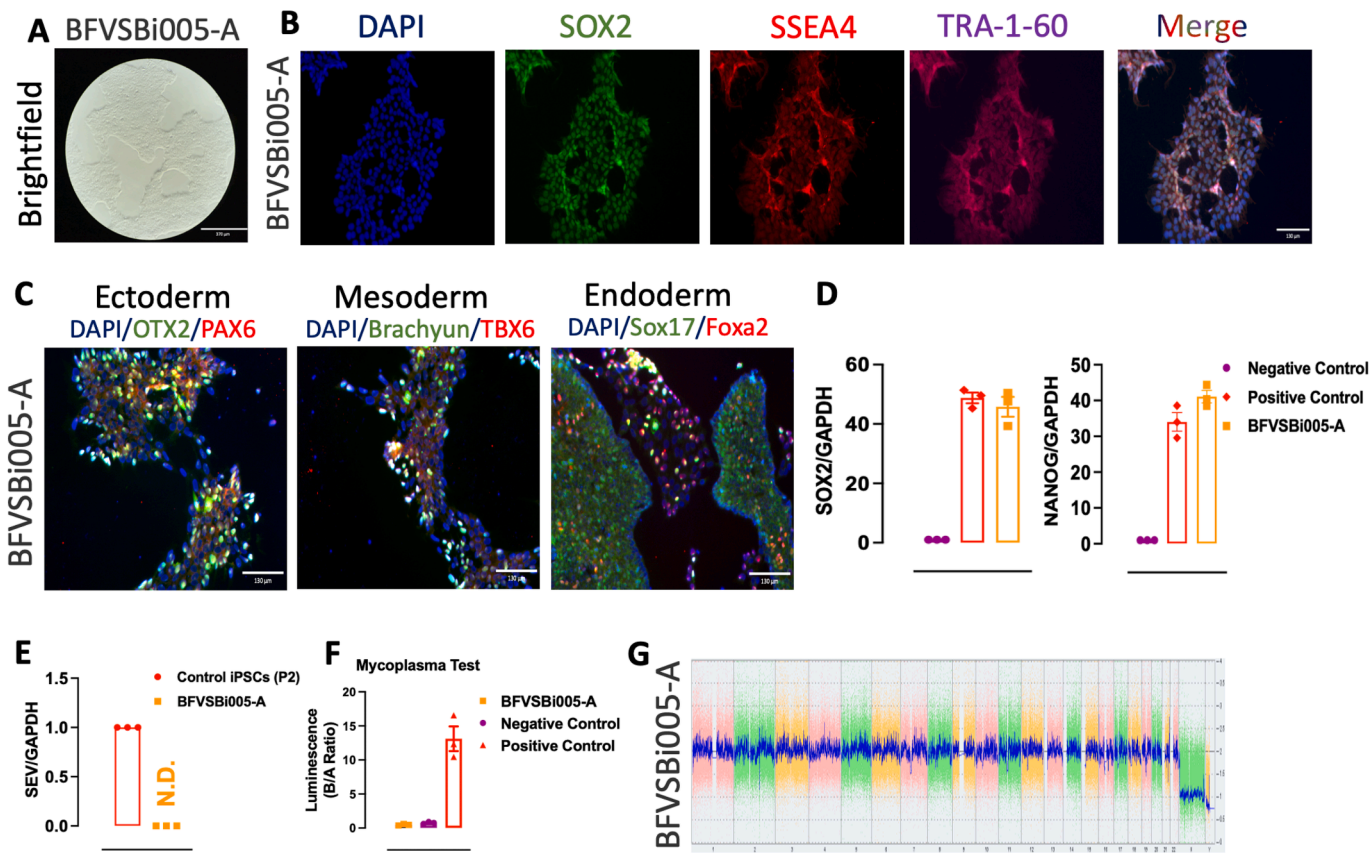

Fig. 1.

effects, with multiorgan dysfunction affecting the lungs, brain, and the cardiovascular system. Notably, persistent endothelial dysfunction in long COVID patients can create a pro-thrombotic state, increasing the risk of coagulation disorders including thromboembolism, which can lead to serious cardiovascular events such as stroke or heart failure. Indeed, a previous study on long COVID have connected endothelial dysfunction to myocardial injury, providing insight into additional systemic complications (Thomas et al., 2024).

We established an induced pluripotent stem cell (iPSC) line from a 64-year-old male (BFVSBi005-A) clinically diagnosed with long COVID, who underwent below-knee amputation following a COVID-19-related thrombotic event. While the exact mechanisms of multi-organ damage in long-term COVID remains unclear, investigating the pathways associated with endothelial dysfunction and its influence on immune response may offer important insights into treatment strategies. Previous research indicates that endothelial and microvascular dysfunction, along with disrupted cellular crosstalk, can worsen organ damage (Sayed et al., 2020). Using iPSCs from this patient as an *in vitro* disease

model allows for the study of cellular interactions and the exploration of underlying mechanisms, potentially aiding in the discovery of new therapeutic options (Sayed et al., 2016).

This iPSC line was created by reprogramming peripheral blood mononuclear cells (PBMCs) with Sendai virus vectors containing the Yamanaka factors (c-Myc, Oct3/4, Klf4, and Sox2). A cloned colony was expanded to passage 14, resulting in iPSCs that showed normal morphology (Fig. 1A) and expressed pluripotency markers (SOX2, SSEA4, and TRA-1–60) (Fig. 1B). A key feature of pluripotency is the ability to differentiate into the three germ layers, verified in this iPSC line by immunostaining for OTX2, SOX17, and BRACHYURY, markers for ectoderm, endoderm, and mesoderm, respectively (Fig. 1C). To further quantify the expression of the pluripotency markers, reverse transcription-quantitative polymerase chain reaction (RT-qPCR) was used to assess NANOG and SOX2 mRNA levels (Fig. 1D). Additionally, RT-qPCR confirmed that the iPSCs were negative for Sendai virus vector mRNA (Fig. 1E). Karyotyping with the KaryoStat assay showed a normal karyotype in this iPSC line (Fig. 1F). Mycoplasma testing confirmed the iPSC line is mycoplasma-free (Fig. 1G), and short tandem repeat (STR) analysis verified the genetic lineage matches that of the donor PBMCs (Submitted in the archive with Journal).

3. Materials and methods

3.1. Reprogramming

PBMC isolation was performed with a percoll density gradient medium (GE Healthcare #17089109) to separate PBMCs from patient blood. PBMCs were then purified with DPBS and cultured as previously described (Tripathi et al., 2024). Culturing media consisted of StemPro®-34 SFM medium supplemented with specific growth factors and cytokines: 100 ng/mL FLT3 (ThermoFisher Scientific, #PHC9414), 100 ng/mL SCF (Peprotech, #300-07), 20 ng/mL IL-3 (Peprotech, #200-3), 20 ng/mL EPO (ThermoFisher Scientific, #PHC9631), and 20 ng/mL IL-6 (ThermoFisher Scientific, #PHC0063) and was used to support PBMCs as previously described (Dexheimer et al., 2024). The CytoTune™-iPSC 2.0 Sendai Reprogramming Kit (#A16517, ThermoFisher Scientific) was used to reprogram PBMCs according to manufacturer instructions. $2x10^5$ PBMCs were plated in one well of a 24-well plate and replated onto Matrigel-coated plates after transduction. Replated PBMCs were cultured in StemPro™-34 medium until day seven post-transduction, and cells were expanded in StemMACS™ iPS-Brew medium (Miltenyi Biotec, #130-104-368). After 10–15 days of culture with StemMACS™ iPS-Brew medium, a colony was picked and clones were expanded as previously described in (Manhas et al., 2024).

3.2. Cell culture

iPSCs were cultured in StemMACS™ iPS-Brew XF media at 37°C with 5% CO_2. Upon reaching 85% confluency, iPSCs were detached using TrypLE™ (ThermoFisher Scientific, #12605010). iPSCs were plated onto Matrigel-coated plates with 10 uM Y-27632 ROCK inhibitor (Selleck Chemicals, #Y27632). Media was refreshed every two days until confluency (3–4 days).

3.3. Trilineage differentiation

The trilineage differentiation kit (STEMCELL™ Technologies, #05110) was used in the differentiation of the iPSC line into the 3 germ layers (mesoderm, ectoderm, endoderm).

3.4. Immunofluorescence

The iPSC-derived germ layers and iPSCs used to visualize expression of pluripotency markers were fixed with 4% paraformaldehyde (10–15 min). The cells were then permeabilized using digitonin (10 min) and blocked at room temperature with Bovine Serum Albumin (Sigma-Aldrich, #A7030) and 5% donkey or goat serum (30 min). After overnight incubation at 4°C with primary antibodies (Table 2), cells were incubated with diluted secondary antibody (Table 2). Cells were imaged after staining with NucBlue (ThermoFisher Scientific, #R37606).

Table 2
Reagents details.

Antibodies used for Immunocytochemistry				
	Antibody	**Dilution**	**Company Cat #**	**RRID**
Pluripotency Markers	Mouse IgG1κ Anti-SOX2	1:200	Santa Cruz Biotechnology Cat# sc-365823	AB_10842165
	Rabbit Anti-SSEA4	1:200	ThermoFischer Scientifc Cat# 620-500	AB_3674317
	Mouse Anti-TRA-1–60 (R)	1:100	Novus Biologicals Cat# NBP2-80985	AB_3410059
Ectoderm Markers	Goat Anti-OTX2	1:200	R&D Systems Cat# 963273	AB_2157172
	Rabbit Anti-Pax6	1:100	Thermo Fisher Scientific Cat# 42-6600	AB_2533534
Endoderm Markers	Goat Anti-SOX17	1:200	R&D Systems Cat# 963121	AB_355060
	Rabbit Anti-Foxa2	1:250	Thermo Fischer Scientific Cat# 701698	AB_2576439
Mesoderm Markers	Goat Anti-Brachyury	1:200	R&D Systems Cat# 963427	AB_2200235
	Rabbit Anti-Tbx6	1:200	Thermo Fischer Scientific cat # PA5-35102	AB_2552412
Secondary Antibodies	Alexa Fluor 488 Goat Anti-Mouse IgG1	1:1000	Thermo Fisher Scientific #A-21121	AB_2535764
	Alexa Fluor 488 Donkey Anti-Goat IgG (H + L)	1:1000	Thermo Fisher Scientific #A-11055	AB_2534102
	Alexa Fluor 555 Goat Anti-Rabbit IgG (H + L)	1:500	Thermo Fisher Scientific #A-21428	AB_141784
	Alexa Fluor 647 Goat Anti-Mouse IgG2b	1:250	Thermo Fisher Scientific #A-21242	AB_2535811

Primers		
	Target	**Forward/Reverse primer (5′-3′)**
Sendai Virus	Sendai Virus genome	Mr04269880_mr
Genotyping	N/A	N/A
House-Keeping Gene	GAPDH	HS02758991_g1
	SOX2	HS01053049_s1
	NANOG	HS02387400_g1

3.5. RT-PCR

At passage 14, iPSCs were collected with TRIzol® and the Directzol™ RNA Microprep Kit (Zymo Research, #R2062) was used to isolate total RNA. cDNA was synthesized with the iScript™ cDNA Synthesis Kit (Bio-Rad, #1708891). The TaqMan™ Gene Expression Assay (Applied Biosystems™, #4444556) and primers (Table 2) were used to quantify expression of NANOG, SOX2, and SEV.

3.6. Karyotyping

At passage 14, 2 x 10^6 iPSCs were collected and the pellet was

analyzed for chromosomal abnormality using the KaryoStat™ assay (ThermoFisher Scientific).

3.7. Short tandem repeat analysis

The QIAamp® DNA Micro Kit (Qiagen, #56304) was used to isolate and purify genomic DNA from PBMCs and iPSCs. The CLA IdentiFiler™ Direct PCR Amplification Kit (ThermoFisher Scientific, #A44660) was used to amplify DNA. The product was analyzed via capillary electrophoresis on an ABI3130xl.

3.8. Mycoplasma detection

The MycoAlert™ PLUS Mycoplasma Detection Kit (Lonza, #LT07-705) was used to test iPSCs for mycoplasma.

CRediT authorship contribution statement

David Wu: Writing – review & editing, Writing – original draft, Data curation. **Amit Manhas:** Writing – review & editing, Methodology, Formal analysis, Data curation, Conceptualization. **Chikage Noishiki:** Data curation. **Dipti Tripathi:** Writing – review & editing, Methodology, Data curation. **Lu Liu:** Data curation. **Naima Turbes:** Data curation. **Dilip Thomas:** Writing – review & editing, Formal analysis, Data curation. **Karim Sallam:** Writing – review & editing, Resources. **Jason T. Lee:** Supervision, Resources, Methodology. **Nazish Sayed:** Writing – review & editing, Funding acquisition.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Amit Manhas reports financial support was provided by American Heart Association Postdoctoral Fellowship. Nazish Sayed reports financial support was provided by National Institutes of Health. Nazish Sayed reports financial support was provided by AHA SFRN. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This study was supported by research grants from the National Institutes of Health (NIH) R01 HL158641, R01 HL161002, and American Heart Association (AHA) SFRN grant 869015 (NS); and AHA Postdoctoral fellowship 23POST1020812 (AM).

Author contributions

All authors contributed significantly to the work presented in this paper. N.S., D.W., and A.M. conceived and designed the experiments; D.W., C.N., L.L. performed the experiments; D.W., N.T., and D.T. analyzed the data; K.S., and J.L. contributed reagents and materials; D.W. wrote the manuscript with input from all authors. All authors have read and approved the final version of the manuscript.

References

Dexheimer, R., Manhas, A., Wu, D., Tripathi, D., Yu Chan, S., Li, J., Yu, R., Sayed, N., Wu, J.C., Sallam, K., 2024. Generation of two iPSC lines from dilated cardiomyopathy patients with pathogenic variants in the SCN5A gene. Stem Cell Res. 80, 103498. https://doi.org/10.1016/j.scr.2024.103498. Epub 2024 Jul 20 PMID: 39067410.

Manhas, A., Tripathi, D., Noishiki, C., Wu, D., Liu, L., Sallam, K., Lee, J.T., Fukaya, E., Sayed, N., 2024. Generation of two iPSC lines from vascular Ehlers-Danlos Syndrome (vEDS) patients carrying a missense mutation in COL3A1 gene. Stem Cell Res. 79. https://doi.org/10.1016/j.scr.2024.103485. Epub 2024 Jun 25. PMID: 38944978; PMCID: PMC11315230.

Sayed, N., Liu, C., Wu, J.C., 2016. Translation of human-induced pluripotent stem cells: from clinical trial in a dish to precision medicine. J. Am. Coll Cardiol. 67 (18), 2161–2176. https://doi.org/10.1016/j.jacc.2016.01.083. PMID: 27151349; PMCID: PMC5086255.

Sayed, N., Liu, C., Ameen, M., Himmati, F., Zhang, J.Z., Khanamiri, S., Moonen, J.R., Wnorowski, A., Cheng, L., Rhee, J.W., Gaddam, S., Wang, K.C., Sallam, K., Boyd, J. H., Woo, Y.J., Rabinovitch, M., Wu, J.C., 2020. Clinical trial in a dish using iPSCs shows lovastatin improves endothelial dysfunction and cellular cross-talk in LMNA cardiomyopathy. Sci Transl Med. 12 (554). https://doi.org/10.1126/scitranslmed.aax9276 eaax9276, PMID: 32727917; PMCID: PMC7557117.

Thomas, D., Noishiki, C., Gaddam, S., Wu, D., Manhas, A., Liu, Y., Tripathi, D., Kathale, N., Adkar, S.S., Garhyan, J., Liu, C., Xu, B., Ross, E.G., Dalman, R.L., Wang, K.C., Oro, A.E., Sallam, K., Lee, J.T., Wu, J.C., Sayed, N., 2024. CCL2-mediated endothelial injury drives cardiac dysfunction in long COVID. Nat. Cardiovasc. Res. 3 (10), 1249–1265. https://doi.org/10.1038/s44161-024-00543-8. Epub 2024 Oct 14 PMID: 39402206.

Tripathi, D., Manhas, A., Noishiki, C., Wu, D., Adkar, S., Sallam, K., Fukaya, E., Leeper, N.J., Sayed, N., 2024. Generation of induced pluripotent stem cell line from a patient suffering from arterial calcification due to deficiency of CD73 (ACDC). Stem Cell Res.